

Using Pyrobot to Control the Lab's Robots

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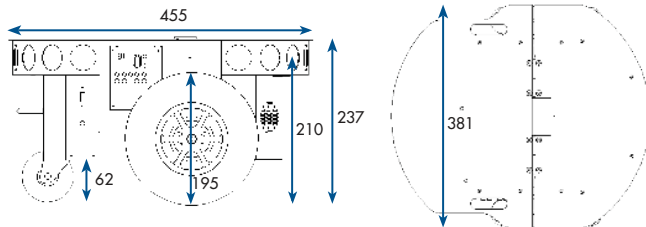
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The Pioneer Robot



The Pioneer Robot

Pioneer 3-DX Dimensions (in mm):



Pioneer 3-DX Specifications:

- differential drive with single rear caster
- odometry: 33,500 counts per motor revolution
- turn radius: 0cm

Pioneer 3-DX Specifications (cont.):

- maximum forward/backward speed: 1.4 m/sec
- rotation speed: 300 deg/sec
- maximum traversable step: 2 cm
- maximum traversable gap: 5 cm
- maximum traversable grade: 25%
- 12v 7.2Ah lead acid sealed batteries (3x)
- weight: 13kg (5.5kg without batteries), payload: 17kg
- 8 sonar: 1 on each side (left and right) and 6 at 20° intervals

The Robot's Computer

On board computer (MSI Cubi-035XEU):

- Ubuntu 24.04
- Intel i3-5005U
- 4GB RAM (DDR3L-SDRAM)
- 128 GB SSD hard drive
- On-board Intel HD Graphics 5500



Image from <https://www.msi.com/Desktop/Cubi/Gallery>

The Robot's Video Camera

Video Camera (Logitech HD Pro Webcam C920)

- Auto Focus
- Focal length: 3.67 mm
- Diagonal Field of View: 78°
- Video Capture: 16:9 W at 360p (640x360), 480p (854x480), 720p (1280x720), and 1080p (1920x1080)
- Video Format: H.264
- Max frame rate: 1080p@30fps
- True optical resolution: 3MP



Image from http://support.logitech.com/en_us/product/hd-pro-webcam-c920

Using the Robots - Practical Considerations

You need to use ssh to connect to the robot's PC.

For example:

```
ssh -X robotica@192.168.1.25
```

```
ssh -X robotica@192.168.1.26
```

Note: '-X' is to give the robot permission to open windows on your machine.

We will use one shared account:

username: robotica password: upmRobotica

Using the Robots - Practical Considerations

Regarding files:

- Use `scp` to copy files to/from the robot. From a terminal on one of the lab's PCs or your laptop:

```
scp myBrain.py robotica@192.168.1.25:myDir  
scp robotica@192.168.1.25:myBrain.py .
```

- Do not forget to delete your work from the robot when finished
- Please don't modify shared files (but if you find a problem with those files let us know).

Pyrobot and the Robots

After starting Pyrobot you need to select:

- Robot: Aria
(No Server!)

or, to save time, you can start things from the command line:

```
pyrobot -r Aria.py -b Avoid.py
```

The Pioneer's Control Panel

The red PWR led is lit when the robot is on.

The BATTERY led is green when fully charged ($> 12.5V$) and changes to orange and then red when discharged ($< 11.5V$).

The red RESET button resets the robot's microcontroller and disables the motors.

The white MOTORS button disables and enables the motors.

A buzz is made when the robot turns on and when a client connects.

Don't push the robot "by hand" when the motors are enabled.



Shutting down the Pioneer

Before turning off the robot please remember to shut down the robot's PC cleanly:

```
sudo shutdown -h now
```

Please leave the robots recharging when you are done working.

You can operate the robot while charging the batteries, but **be careful** with the wire when moving the robot.

The Lab's Network

When working in the lab area the robots automatically connect to a private WIFI network. The desktop computers in the lab are also connected to that same network.

If you wish to use your laptop to control the robots you also need to connect to that same network:

- SSID: DIA-ROBOTICA
- Password: upmRobotica71655

Robot and PC IP addresses are given in stickers on each machine.

Using the Robot, Step By Step (1/2)

- Connect your laptop to the lab's network (DIA-ROBOTICA)
- Turn on the robot's base, then turn on the robot's PC:



- Login to the robot
 - `ssh -X robotica@192.168.1.25`
or
`ssh -X robotica@192.168.1.26`
 - password upmRobotica,
- Copy your files to the robot (see above)
- (Next page)

Using the Robot, Step By Step (2/2)

- Startup pyrobot (`pyrobot -r Aria.py -b YourBrain.py`)
- Work...
- Erase your files.
- Shutdown PC (`sudo shutdown -h now`), Shutoff the Robot's base
- Leave the robot charging

Charge Robot

